

Positive Square Energy of Every Rank-Ten Cactus

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Abstract

Let G be a connected cactus of cyclomatic rank ten on n vertices. We prove $s^+(G) > n$. The sharp block-additive doubly nonnegative estimate leaves exactly T^9Q and T^8PP , where $T = C_3$, $P = C_5$, and Q is an arbitrary cycle. In the disconnected shared-cut case, exact colored-partition audits give $96 = 92 + 4$ and $180 = 170 + 10$ proper rows. The only endpoint kernels have the exact closures $A_9 \mid Q$: $3624 = 3618 + 6$, entry-locked $T^8P \mid P$: $11689 = 11586 + 100 + 3$, and $P \mid A_8 \mid P$: $11689 = 11674 + 15$. In the fully shared case the complete stabilized T^9Q census has 8049 types and three hostile exceptions. The complete T^8PP census is $30386 = 30377 + 9$; all nine replacements are verified, including all $5 \cdot 4 \cdot 3 = 60$ cyclic placements in the pentagon-router case N4. Global cuts are actual bridges, local cuts are proper cycle intervals, later cuts refine one owner, and arbitrary attached trees follow their unique final hull owner. The proof uses neither a separator theorem nor a two-pivot phase or winding assertion.

1 Definitions, inputs, and proof firewall

All graphs are finite, simple, and undirected. A cactus is a connected graph whose blocks are bridges or cycles; its cyclomatic rank is the number of cyclic blocks. If the adjacency eigenvalues of G are $\lambda_1, \dots, \lambda_n$, put

$$s^+(G) = \sum_{\lambda_i > 0} \lambda_i^2, \quad s^-(G) = \sum_{\lambda_i < 0} \lambda_i^2, \quad \sigma(G) = s^+(G) - |V(G)|.$$

For an induced vertex partition $V(G) = V_1 \dot{\cup} \dots \dot{\cup} V_r$, positive square energy is superadditive [1]:

$$s^+(G) \geq \sum_i s^+(G[V_i]), \quad \sigma(G) \geq \sum_i \sigma(G[V_i]). \quad (1)$$

A connected rank-ten graph has $|E| = n + 9$, so the main theorem proves strictly the conjectured lower bound of [2] for this class.

Write $T = C_3$, $P = C_5$, and

$$\delta = \sec(\pi/5) - 1 = \sqrt{5} - 2 < \frac{1}{4}, \quad \delta_q = \sec(\pi/q) - 1 < 1.$$

A cycle $Q = C_q$ is *hostile* when $q \equiv 1 \pmod{4}$.

The dependency boundary is the corrected synthesis [7], accepted by the hostile audit [8]. We use the authoritative corrected octacyclic infrastructure [5], the proved rank-nine theorem [6], the exact sharp-DNN frontier [9], and the finite certificates cited at each census. Those references are proof packets, not numerical evidence: they supply the analytic inequalities or finite exhaustive checks stated here.

Two statements are expressly outside the proof. We do not use the open two-pivot Schur–Sachs phase/winding assertion [22]; its exact two-terminal determinant state does not imply the required integrated phase bound. We do not use candidate separator Lemma S or any census-free “good-router” assertion [23]. No connector is contracted, no edge-monotonicity or principal-arctan shortcut is used, and no two-interface row is inferred from an unproved Voronoi rule.

2 Packet bounds, intervals, and ownership

The rank-two through rank-nine cactus theorems and the corrected octacyclic packet infrastructure give, uniformly over arbitrary finite trees attached at arbitrary packet vertices,

$$\sigma(H) \geq 0 \quad \text{for connected ranks two and three,} \quad (2)$$

$$\sigma(H) > 0 \quad \text{for connected ranks four through nine,} \quad (3)$$

$$\sigma(P) \geq -\delta, \quad \sigma(Q) \geq -\delta_q \quad \text{for hostile } Q, \quad (4)$$

$$\sigma(TP) > \frac{3}{4}, \quad \sigma(TQ) > 0, \quad (5)$$

$$\sigma(PP) > 0, \quad \sigma(TPP) > \frac{3}{2} \quad \text{for one shared-cut cluster.} \quad (6)$$

The strict PP estimate in (6) has precisely the displayed scope: the two pentagons belong to one connected shared-cut cluster, with arbitrary finite trees attached at its vertices. This is the corrected packet theorem of [5, Eq. (2.5)]; no claim is made here for an arbitrary two-pentagon cactus joined by a positive-length bridge. Here and below a packet name permits arbitrary joining paths and rooted trees within the scope of the cited packet theorem. For one connected shared-cut cluster A_r of r triangles,

$$\sigma(A_r) > b_r, \quad (b_1, \dots, b_9) = (0, 1, 2, 3, 2, 1, 0, 0, 0). \quad (7)$$

The entries through $r = 8$ are the corrected triangular-cluster bounds in [5]; $r = 9$ uses [6]. Only strict positivity is used at $r = 8, 9$.

Two analytic packets close the locked one-pivot rows.

Lemma 2.1 (Rooted packing one). *Let a connected cactus H contain one hostile cycle $Q = C_q$ and $a \geq 1$ triangles, no two vertex-disjoint. The cycle Q may meet their common root or be joined to it by an arbitrary path. Arbitrary trees may be attached to the cycles, joining path, and its vertices. Then*

$$s^+(H) - s^-(H) > -2\delta_q, \quad \sigma(H) > a - \delta_q.$$

The same statement with $Q = P$ gives $\sigma(H) > a - \delta$.

Proof. This is the rooted packing-one theorem of [19]. Briefly, exact tree elimination replaces each spine activity by a positive matching message $\alpha_v \geq t$. Grouping the Sachs expansion by the unique hostile cycle gives $\Psi_H(t) = R + 2i(B - A)$ with $R > 0$. Packing one excludes products of two triangle terms, and splitting matchings by boundary use gives

$$R - Z_q(t)(B - A) = E + LB + Z_q(t)A + 4 \sum_{T \cap Q = \emptyset} Z_{H-(V(Q) \cup V(T))}(\alpha) > 0.$$

Thus the continuous argument is strictly smaller than the isolated hostile cycle argument. The signed Coulson identity gives the first inequality, and $s^+ + s^- = 2|E| = 2(|V| + a)$ gives the second. The message proof already includes arbitrary rooted trees and arbitrary joining-path length. $\square$

Lemma 2.2 (Common-cut Schur–Sachs packets). *If all named cycles share one actual cut vertex, uniformly over arbitrary attached trees,*

$$\sigma(T^k Q) > k - \delta_q \quad (q \equiv 1 \pmod{4}), \quad (8)$$

$$\sigma(T^k Q) > k \quad (q \text{ even or } q \equiv 3 \pmod{4}), \quad (9)$$

$$\sigma(T^k P) > k - \delta, \quad (10)$$

$$\sigma(T^k PP) > k + 1 - \frac{4}{3\sqrt{13}}. \quad (11)$$

Proof. After exact tree elimination, the normalized determinant is the scalar state

$$W(t) = a_x + \sum_j \frac{B_j + \omega_{\ell_j}}{A_j}, \quad \omega_{\ell} = -2i^{-\ell},$$

with $A_j, B_j, a_x > 0$. The one-hostile-cycle phase is bounded strictly by the isolated-cycle phase. For two pentagons the exact coefficient inequality $2R \geq t(t^4 + 7t^2 + 9)(A_1 + A_2)$ and $\int_0^\infty 4/(t^4 + 7t^2 + 9) dt = 2\pi/(3\sqrt{13})$ give the final line. The full argument and its 1290-term exact coefficient certificate are [20, 21]. This scalar lemma is invoked only at one real common pivot; it is not a two-pivot estimate. $\square$

We now state explicitly the graph-realization facts that connect the finite incidence certificates to induced partitions.

Lemma 2.3 (Proper cycle intervals). *Let C be a cyclic block carrying two or three distinct marked vertices. For two marks, one mark and the complementary two-vertex interval on a triangle, or either nonempty proper consecutive arc on a longer cycle, are connected induced intervals. Three marks on a triangle give three singleton intervals. For the N_4 pentagon router, isolating one marked vertex leaves a four-vertex induced path containing the other two marks, for every injection of the three marks into the five cycle vertices. Incidence branches attached at a mark may be assigned to that mark's interval. If a later router lies in one interval, splitting it refines only that interval and preserves an induced partition.*

Proof. Deleting the boundary edges between consecutive arcs of a chordless cycle leaves induced paths (or singletons). Distinct triangle marks have cyclic gaps $1 + 2$ or $1 + 1 + 1$. On C_5 , deleting the isolated marked vertex leaves the unique complementary four-vertex path, independent of the cyclic placement of the other two marks. Attached cactus branches meet the cycle at only their marked cut, so adjoining a branch preserves connectedness and inducedness. Refinement replaces one part by an induced partition of that part. $\square$

Lemma 2.4 (Final-owner realization). *Contract maximal shared-cut cyclic clusters while retaining every actual bridge and every internal connector vertex. The resulting cluster graph is a tree. A partition of it into connected subtrees, together with legal interval splits from Lemma 2.3, realizes connected, induced, disjoint, exhaustive territories in G . Every shared cut, connector remnant, incidence branch, cycle interval, and off-hull tree has exactly one final owner. A later split replaces the provisional owner of one active territory; no provisional owner may survive. A private interval containing no retained cycle is a nonempty tree and has exact surplus -1 .*

Proof. The block-cut graph of a cactus is a tree. Contracting connected subtrees of cyclic blocks cannot create a cycle, and retaining bridge paths preserves all vertices. Cutting an actual bridge separates two connected induced vertex sets. At a cyclic router, Lemma 2.3 assigns each vertex and

each attached incidence branch once. Induct on the sequence of refinements; the active territory being split is replaced by disjoint exhaustive induced subterritories, while all other owners remain unchanged. An off-hull tree has a unique first hull attachment and follows its final owner. Finally, for a nonempty tree E , bipartite spectral symmetry and $|E(E)| = |V(E)| - 1$ give $s^+(E) = |V(E)| - 1$ and hence $\sigma(E) = -1$. $\square$

These lemmas are the formal ownership and interval interface of the corrected octacyclic infrastructure [5]. They cover arbitrary connector lengths, branching connector trees, coincident interfaces, and arbitrary finite trees at every core or connector vertex.

3 Exact sharp-DNN reduction

For an edge-containing graph define

$$\kappa(G) = \sup_{\substack{M \succeq 0, \ M \geq 0 \\ \mathbf{1}^T M \mathbf{1} > 0}} \frac{4(\sum_{uv \in E(G)} \sqrt{M_{uv}})^2}{\mathbf{1}^T M \mathbf{1}}.$$

The sharp cactus DNN theorem [4, 3] says

$$\kappa(G) = b + \sum_i \kappa(C_{\ell_i}), \quad \kappa(C_\ell) = \begin{cases} \ell, & \ell \text{ even,} \\ \ell \sec^2(\pi/(2\ell)), & \ell \text{ odd,} \end{cases} \quad s^-(G) \leq \kappa(G).$$

Put $\epsilon_\ell = 0$ for even ℓ and $\epsilon_\ell = \ell \tan^2(\pi/(2\ell))$ for odd ℓ . If the ten cyclic blocks have lengths $\ell_1, \dots, \ell_{10}$, block counting and $|E| = n + 9$ give the exact bound

$$\sigma(G) \geq 9 - \sum_{i=1}^{10} \epsilon_{\ell_i}. \quad (12)$$

The odd sequence decreases strictly. With $a = \epsilon_5 = 5 - 2\sqrt{5}$,

$$\epsilon_3 = 1, \quad 3a < 2, \quad 2a > 1, \quad \epsilon_5 + \epsilon_7 < 1.$$

The first two comparisons reduce after squaring positive sides to $169 < 180$ and $80 < 81$; the last follows from $\epsilon_7 < 7/15 < 2\sqrt{5} - 4$, with $4489 < 4500$. These exact certificates and monotonicity are recorded in [9].

If at most seven cycles are triangles, then $\sum \epsilon_i \leq 7 + 3a < 9$. With exactly eight triangles, the two remaining cycles contribute at least one only when both are pentagons: an even cycle contributes zero, and every other odd pair contributes at most $\epsilon_5 + \epsilon_7 < 1$. At least nine triangles has the form $T^9 Q$, including $Q = T$. Consequently the exact, disjoint DNN residual frontier is

$$T^9 Q = \{3, 3, 3, 3, 3, 3, 3, 3, 3, q\}, \quad q \geq 3, \quad T^8 PP = \{3, 3, 3, 3, 3, 3, 3, 3, 5, 5\}. \quad (13)$$

Every other rank-ten cycle multiset is strict directly from (12).

4 Disconnected shared-cut clusters

For a residual graph the reduced shared-cut cluster tree either has one node or more than one. In the latter case the exact colored integer-partition audit [10] is

residual	all partitions	proper	direct packet rows	structural rows
T^9Q	97	96	92	4
T^8PP	181	180	170	10

The one-cluster partition is excluded. Every direct row has a positive exact sum of (2)–(7), or a zero sum containing a strict packet.

For T^9Q the four frozen structural rows are

$$Q + T + \cdots + T, \quad Q + T + T + A_7, \quad Q + T + A_8, \quad Q + A_9.$$

A singleton-triangle leaf leaves a strict rank-nine complement. If no such leaf exists in $Q + T + T + A_7$, the four nodes form a path with Q, A_7 at its ends; the singleton next to Q forms strict TQ , and the other with A_7 is a strict rank-eight cactus. In $Q + T + A_8$, either T is a leaf or the middle node and forms TQ , leaving strict A_8 . The unique endpoint is

$$A_9 \mid Q. \tag{14}$$

For T^8PP the ten rows are

$$P + P + T + \cdots + T, \quad P + P + T + A_7, \quad P + P + A_8,$$

and the seven rows $P + T + \cdots + T + (T^kP)$, $2 \leq k \leq 8$, with the singleton count completing eight triangles. Remove any singleton-triangle leaf. If none exists in the first row, the pentagons are the leaves of a path; a terminal TP leaves a strict rank-eight complement. In $P + P + T + A_7$, a leaf A_7 leaves strict TPP , and otherwise a terminal TP leaves strict rank-eight A_7P . In $P + P + A_8$, a middle pentagon gives $PP + A_8$; only $P \mid A_8 \mid P$ remains. For $P + T + \cdots + T + (T^kP)$ with $k < 8$, the bare pentagon and nearest singleton give TP and a strict rank-eight remainder. At $k = 8$ there is no singleton. Thus the only endpoints are

$$T^8P_0 \mid P_1, \quad P_0 \mid A_8 \mid P_1. \tag{15}$$

Every move here is a leaf/path move in the actual reduced tree; no separator claim is being made.

4.1 The endpoint $A_9 \mid Q$: $3624 = 3618 + 6$

Mark the first cyclic-hull entry of the actual connector from Q into A_9 . An entry is a shared cut or an actual private triangle vertex; an off-hull entry projects to its unique hull attachment. The exact census [11, 12] contains

unmarked A_9 incidence trees	355
marked placements before automorphisms	6745
canonical marked rows	3624
accepted finite-router rows	3618
explicit repairs	6.

For an accepted row the checker verifies proper intervals, disjoint exhaustive branches, and one final connector owner. If the Q territory retains k triangles, then $k = 1$ uses strict TQ ; $k = 2$ uses nonnegative TTQ plus a strict branch; $3 \leq k \leq 8$ uses the lower-rank theorem; and for $k = 0$ the other eight triangles occupy at most two branches, one of size at least four and margin > 3 , absorbing $Q > -1$.

The six exceptions are the hub/private entry orbits of the common-cut bouquet and four entry orbits on one two-cut saturated extension. The bouquets have packing one and margin $> 9 - \delta_q$. In one extension orbit the leaf triangle and Q form strict TQ , leaving strict common-cut A_8 . In the other three, open the leaf triangle: its private territory is a nonempty tree of surplus -1 , while the eight retained hub triangles and actual path to Q satisfy Lemma 2.1. Hence

$$\sigma(G) > (8 - \delta_q) - 1 = 7 - \delta_q > 0.$$

This closes exactly $3624 = 3618 + 6$ rows. Nonhostile Q , including $Q = T$, has the same or stronger ledger.

4.2 Entry-locked $T^8P \mid P$: $11689 = 11586 + 100 + 3$

An internal clustered pentagon and a privately entered incidence-leaf pentagon are handled directly by proper pentagon intervals. The remaining entry-locked family [13] has 2392 colored T^8P incidence trees, 1105 with clustered P an incidence leaf, and 11689 canonical marked-entry rows. Exact final-owner certificates close 11586 directly and 100 by finite replacement sequences. Those replacements use zero, one, two, and three routers in respectively 2, 9, 73, 16 rows. The verifier checks every split order, interval, active territory, retained component, packet hypothesis, radical ledger, connector/root/attachment owner, and rejects every completed state with a provisional **territory:*** owner.

The remaining three marked orbits lie on one two-cut shape: a router triangle joins the clustered leaf pentagon to a common-hub fan of eight triangles. Open the remote P_1 . Its four private vertices and all trees rooted there form a nonempty tree of surplus -1 ; the retained packet owns all eight hub triangles, P_0 , both actual connector remnants, and their trees. Packing one gives $> 8 - \delta$, and therefore

$$\sigma(G) > (8 - \delta) - 1 = 7 - \delta > 0.$$

Thus the exact closure is $11689 = 11586 + 100 + 3$.

4.3 The endpoint $P \mid A_8 \mid P$: $11689 = 11674 + 15$

The two labelled entries range independently over every shared cut and actual private triangle vertex, including coincidence. The exact census [14, 15] has 126 unmarked incidence trees, 36414 ordered labelled placements, 11689 canonical rows, 11674 accepted router rows, and 15 explicit repairs. Every accepted row has integer credit at least one after private interval charges, so its two pentagonal deficits leave $> 1 - 2\delta = 5 - 2\sqrt{5} > 0$. The checker assigns both connectors and both remote pentagons once and prevents a destroyed router triangle from reappearing.

Six exceptions are bouquet mark orbits. Their repairs are: hub–hub or the suitable private profiles, open one remote pentagon and retain packing-one T^8P , giving $> 7 - \delta$; hub–private, split to $P+$ common-cut T^7P , giving $> 7 - 2\delta$; coincident private entries, strict $A_7 + PP$; and the remaining private profiles, the same opening or proper triangle intervals. The other nine lie on a saturated router between a singleton triangle and a six-triangle common-cut branch. Its three singleton ports give, according to orbit,

$$A_6 + T + PP > 1, \quad A_6 + TP + P > 2 - 2\delta,$$

or $T^6P + T + P > 6 - 2\delta$, using respectively a real common cut or packing one. The materialized repairs distinguish TP and PP terminals and check all final owners. Hence all $11689 = 11674 + 15$ rows close.

5 One fully shared cluster

5.1 Complete T^9Q census and exceptions

The exact color-preserving incidence census [16, 12], by shared-cut number c , is

Q capacity	1	2	3	4	5	6	7	8	9	total
$q = 3$	1	12	91	406	1178	2115	2250	1246	275	7574
$q = 4$	1	12	91	412	1203	2187	2361	1340	306	7913
$q = 5$	1	12	91	412	1208	2201	2393	1372	321	8011
$q = 6$	1	12	91	412	1208	2204	2400	1383	327	8038
$q = 7$	1	12	91	412	1208	2204	2402	1386	330	8046
$q = 8$	1	12	91	412	1208	2204	2402	1387	331	8048
$q = 9$	1	12	91	412	1208	2204	2402	1387	332	8049
$q \geq 10$	1	12	91	412	1208	2204	2402	1387	332	8049

The universe stabilizes because Q can meet at most nine other cycle nodes. Ordinary exact splits close every nonhostile row except the common-cut bouquet. In each hostile ledger regime they leave exactly three rows:

1. all nine triangles and Q share one cut; Lemma 2.2 gives $> 9 - \delta_q$;
2. all nine triangles contain one hub and Q joins through a router triangle; Lemma 2.1 gives $> 9 - \delta_q$ along the actual joining segment;
3. a unique leaf triangle lies opposite an eight-triangle common-hub lobe; opening it costs -1 , and packing one on the retained lobe and complete path to Q gives $> 7 - \delta_q$ in total.

At $q = 5$ there are 8011 types and at saturated capacity nine there are 8049; the closure verifier reruns every ordinary split in both hostile regimes, matches all three frozen signatures, and checks cycle coverage, router intervals, and unique retained-cut ownership. Nonhostile ledgers are stronger and leave only the first row. This is the complete fully shared T^9Q census, not a stabilized extrapolation.

5.2 Complete T^8PP census: $30386 = 30377 + 9$

The exact census is

shared cuts c	1	2	3	4	5	6	7	8	9	total
all	1	19	204	1155	3990	8135	9615	5843	1424	30386
ordinary safe	0	17	200	1154	3989	8135	9615	5843	1424	30377
exceptions	1	2	4	1	1	0	0	0	0	9

Every ordinary-safe row has a checked proper-interval split, positive exact packet ledger, and complete final-owner assignment. The nine exceptions and their exact replacements [17, 18] are

code	replacement	margin
N1	common-cut T^8PP	$> 9 - 4/(3\sqrt{13})$
N2	open leaf P ; retain common-cut T^8P	$> 7 - \delta$
N3	$P+$ common-cut T^7P through one router	$> 7 - 2\delta$
N4	$A_7 + TP$ through a pentagon router	$> 3/4$
N5	open one P ; retain packing-one T^8P	$> 7 - \delta$
N6	$P + T+$ common-cut T^6P	$> 6 - 2\delta$
N7	$P + P + A_6$ through two routers	$> 1 - 2\delta$
N8	$P + P + T + A_5$ through two routers	$> 2 - 2\delta$
N9	$P + P + T + T + A_4$ through two routers	$> 3 - 2\delta$

For exact identification their canonical signatures are

N1 X(P)P(T)T(T)T(T)T(T)T(T)T(T))

N2 P(X(P))X(T)T(T)T(T)T(T)T(T)T(T))

N3 T(X(P))X(P)T(T)T(T)T(T)T(T)T(T))

N4 P(X(P))X(T))X(T)T(T)T(T)T(T)T(T)T(T))

N5 T(X(P))X(P))X(T)T(T)T(T)T(T)T(T)T(T))

N6 T(X(P))X(P)T(T)T(T)T(T)T(T)T(T))X(T))

N7 X(T)T(T)T(T)T(T)T(T)T(X(P)))T(X(P)))

N8 X(T)T(T)T(T)T(T)T(X(P)))T(X(P))X(T)))

N9 X(T)T(T)T(T)T(T)X(P))X(T)))T(X(P))X(T)))

N4 is a genuine pentagon interval argument, not a cyclic-order assumption. The verifier materializes all $5 \cdot 4 \cdot 3 = 60$ injections of its three actual marks into the router pentagon. The vertex carrying the A_7 cut is isolated; the complementary four-vertex path contains the other two marks and owns the TP connector packet. It checks the actual connector edges and both packet owners in every placement. In N7–N9 the second triangle split refines one active territory created by the first. In N2 and N5 exactly the four private vertices of the opened pentagon and trees rooted there incur the -1 charge; the entry cut and connector remnants remain with the retained packet.

The replacement verifier regenerates all 30386 incidence types and the exact nine exceptions. It checks frozen signatures and labelled edges, router activity and nesting, proper interval sizes, all N4 cyclic placements, packet connectivity, common-cut or packing-one hypotheses, cycle coverage, connector edges, unique cut/cycle-vertex/attachment ownership, and exact radical signs. Thus $30386 = 30377 + 9$ is a complete closure.

6 Main theorem and exhaustion

Theorem 6.1. *Every connected cactus G of cyclomatic rank ten, on n vertices, satisfies*

$$\boxed{s^+(G) > n}.$$

Proof. The exact DNN inequality (12) proves every cycle multiset except the two disjoint residual families (13). For a residual cactus, the reduced shared-cut cluster tree has one node or more than one.

With more than one node, the exact audits contain 96 proper T^9Q rows, partitioned as $96 = 92 + 4$, and 180 proper T^8PP rows, partitioned as $180 = 170 + 10$. The direct rows have valid packet ledgers. Actual reduced-tree leaf/path moves reduce the structural rows exactly to $A_9 \mid Q$, $T^8P \mid P$, and $P \mid A_8 \mid P$. Their exhaustive marked closures are respectively $3624 = 3618 + 6$, $11689 = 11586 + 100 + 3$, and $11689 = 11674 + 15$.

With one node, the complete capacity-dependent T^9Q census and its three hostile replacements close every incidence. The complete T^8PP census has $30386 = 30377 + 9$ rows, and the nine listed replacements close every exception, including all N4 placements.

Lemma 2.4 upgrades each incidence certificate to connected induced territories. Every global cut is an actual bridge; every local split is a nonempty proper cycle interval; every later split refines one owner; and every shared cut, connector remnant, incidence branch, interval, opened vertex, and attached tree is assigned once. Common-cut bounds are used only at one actual pivot and packing one only when all retained triangles contain the displayed hub. Each opening records its exact nonempty-tree cost. No separator or two-pivot assertion occurs. Equation (1) therefore gives $\sigma(G) > 0$ in every case. $\square$

A Exact reproduction and fail-closed boundary

The census and closure programs require Python 3.10 or newer and use only the standard library. The separate coefficient certificate uses `sympy==1.14.0`. From the repository root, create a self-contained environment and run:

```
python3 -m venv .venv-rank-ten
. .venv-rank-ten/bin/activate
python -m pip install --upgrade pip
python -m pip install sympy==1.14.0
python -c 'import sys; sys.exit("Python 3.10+ required") \
    if sys.version_info < (3, 10) else None'
python research/rank-ten-cactus-residual-partition-audit.py
python -O research/rank-ten-cactus-residual-partition-audit.py
python research/rank-ten-a9-one-interface-census.py
python -O research/rank-ten-a9-one-interface-census.py
python research/rank-ten-t9q-template-closure-verifier.py
python -O research/rank-ten-t9q-template-closure-verifier.py
python research/rank-ten-t8p-entry-locked-census.py
python -O research/rank-ten-t8p-entry-locked-census.py
python research/rank-ten-a8-two-interface-census.py
python -O research/rank-ten-a8-two-interface-census.py
python research/rank-ten-fully-shared-incidence-census.py
python research/rank-ten-t8pp-nine-exceptions-resolution.py
python -O research/rank-ten-t8pp-nine-exceptions-resolution.py
python positive-square-energy/experiments/\
    c5_bouquet_matching_certificate.py
```

The partition audit verifies 97/96/92/4 and 181/180/170/10 and freezes all structural rows. The hardened endpoint and replacement verifiers reproduce $3624 = 3618 + 6$, both hostile T^9Q universes and their three exceptions, $11689 = 11586 + 100 + 3$, $11689 = 11674 + 15$, and $30386 = 30377 + 9$. They use explicit exception-raising checks and pass under `python -O`; classifications use integers

or `Fraction`, and radical signs use exact squared comparisons. The standalone compressed shared-incidence generator still contains Python `assert` statements and is therefore run only without `-O`. The dedicated closure verifiers recheck their stated totals, signatures, ownership conditions, and replacements, but they import census modules and reuse recurrence code. They therefore are not an independent all-row implementation. Pending such a verifier, the computational claim is limited to reproducible, fail-closed checks within this shared code base. The programs certify finite classifications and routing; the analytic packet inequalities are Lemmas 2.1 and 2.2 and the cited lower-rank manuscripts.

Build the paper with

```
bash scripts/build-paper.sh all-rank-ten-cacti
```

or directly with

```
latexmk -pdf -interaction=nonstopmode -halt-on-error \
-cd all-rank-ten-cacti/paper.tex
```

References

- [1] S. Akbari, H. Kumar, B. Mohar, and S. Pragada, *A linear lower bound for the square energy of graphs*, Electron. J. Combin. 32(3) (2025), Paper P3.53.
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- [3] Y. Liu, Q. Tang, and S. Zhang, *The positive and negative square-energy conjecture*, arXiv:2607.18031, 2026.
- [4] Companion manuscript, *The Optimized Liu–Tang–Zhang Constant for Cactus Graphs*, 2026; `sharp-cactus-dnn/paper.tex`.
- [5] Companion manuscript, *Positive Square Energy of Every Octacyclic Cactus*, authoritative corrected infrastructure, 26 July 2026; `all-octacyclic-cacti/paper.tex`.
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